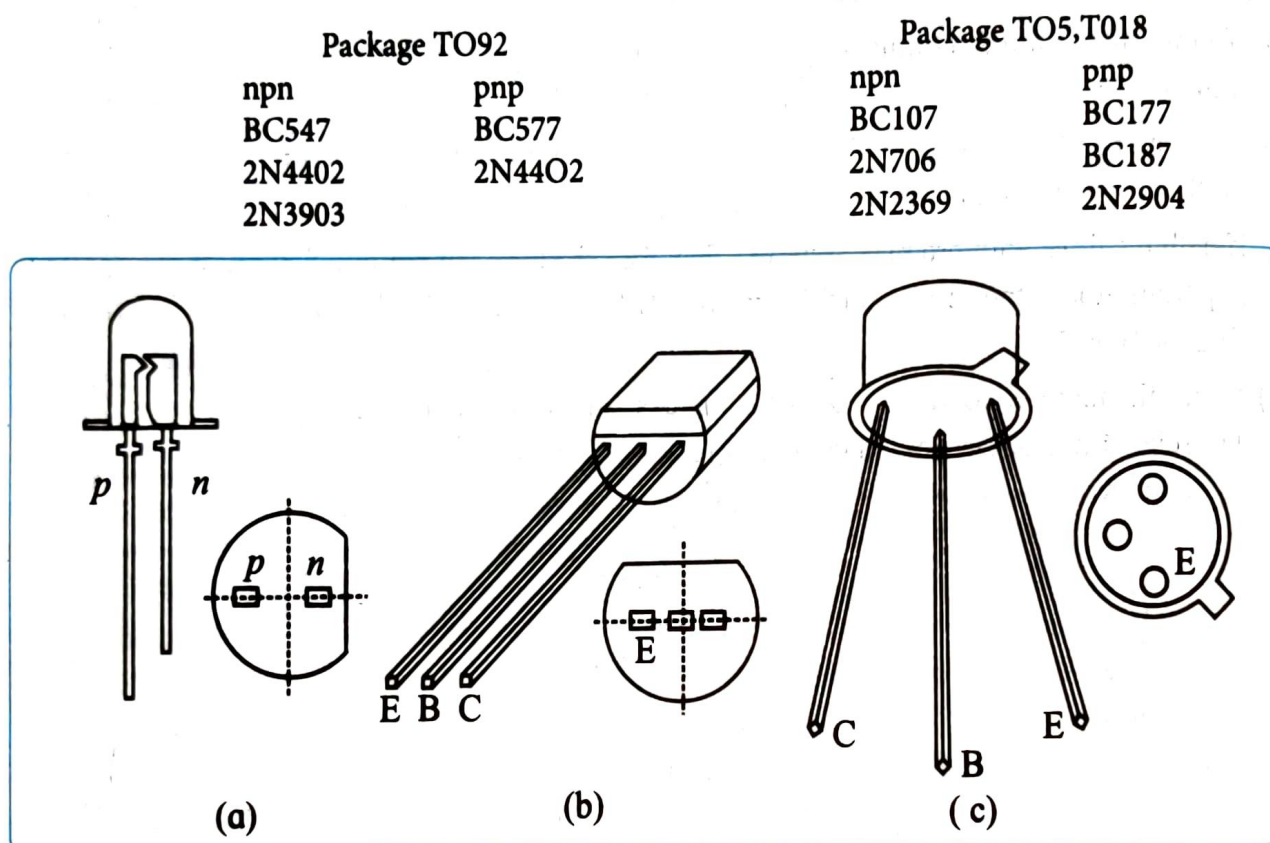


ACTIVITY NO. 8 USE OF MULTIMETER

Aim : Use of multimeter to check working of various components.

Apparatus : Multimeter, transistor (npn and pnp), diode, LED.

Diagram :



Procedure :

i) To identify base of a transistor

ii) To distinguish between npn and pnp type transistors.

1. Make sure that red (positive) and black (negative) probes of DMM are in proper positions.
2. Select the diode-check range of DMM.

Terminology used

- a) **Forward Bias :** When positive of DMM is connected to p-terminal and negative of DMM is connected to n-terminal of a component, with selection of diode-check range of DMM, the reading shown in DMM will be between 0.3 V and 0.8 V, indicating forward bias.
- b) **Reverse Bias :** When positive of DMM is connected to n-terminal and negative of DMM is connected to p-terminal of a component with selection of diode-check range of DMM, there will be no reading shown in DMM. It will indicate open circuit by vertical line (J) which is reverse bias.
- c) **npn transistor :** When positive probe of DMM is connected to base terminal and negative probe of DMM is connected to either collector terminal or emitter terminal, it will show forward bias.

- d) **npn transistor** : when negative probe of DMM is connected to base terminal and positive probe of DMM is connected to either collector terminal or emitter terminal, it will show forward bias.
- 3) Connect the positive probe of DMM to the centre terminal of any transistor (npn or npn)
 - 4) Connect the negative probe in turn to other two terminals. If both the times forward bias shown, it is an npn transistor and the centre terminal is 'base'
 - 5) If the above condition is not satisfied, repeat with other two terminals until you find the 'base'
 - 6) If none of the three terminals satisfy the test for 'base' terminal, then this may be a pnp transistor.
 - 7) Connect the negative probe of DMM to the centre terminal of the transistor and positive probe of DMM to remaining terminals in turn.
 - 8) If both the times forward bias is shown, then the selected centre terminal is 'base' and it is the pnp transistor. If the above condition is not satisfied, repeat with other two terminals, until you find the 'base'.
- iii) To see the unidirectional flow of current in case of a diode and an LED.
- 1) Select the diode-check range of DMM.
 - 2) Connect the two probes of DMM to two terminals of a diode/an LED.

In case of a diode -

When the negative probe of DMM is connected to the cathode (i.e. (n) terminal adjacent to ring marked on diode) it will show forward bias. Thus current flows through diode. When the DMM probes are interchanged, it will show reverse bias or open circuit. That is, no current flows through diode. Thus there is unidirectional flow of current in a diode.

In case of an LED-

When positive probe of DMM is connected to anode (long terminal of LED) and negative probe of DMM is connected to cathode (short terminal of LED) DMM will show forward bias. LED lights up indicating flow of current.

When DMM probes are interchanged, DMM will show reverse bias. LED does not light up, indicating no flow of current. Thus, there is unidirectional flow of current in LED.

- iv) To check whether a given electronic component (diode, transistor) is in working order.

1. Test the component using the above procedure.
2. Diode/LED must satisfy both forward bias and reverse bias tests, it is in working order.
3. In case of a transistor, 'base' identification test should be checked, so that it is in working order. If 'base' is not identified, then transistor is not working order.

Observation Table :

Sr. No	Transistor	Forward Bias		Reverse Bias		Type of Transistor
		CB	BE	CB	BE	
1	DMM	0.3	0.8			nnp Transistor
2	DMM					Pnp Transistor

Result :

1. Transistor is type
2. Transistor is type
3. Diode is type

Precautions:

1. Turn the function knob of the DMM/Multimeter gently so as not to wear out its contact points inside.
2. Switch off the Multimeter. (i.e. turn its function knob to the OFF position) when not using.

Questions

1. Which type of error is present in multimeter?

The error present in a multimeter is generally an instrumental error, which arises due to imperfections in the instrument, calibration errors, or limitations of its accuracy.

2. Can we distinguish between zener diode and rectifier diode using multimeter?

Yes. Using the diode-testing mode of a multimeter, both diodes show conduction in the forward direction and high resistance in the reverse direction. However, to identify a Zener diode conclusively, its reverse breakdown voltage must be measured using a suitable circuit. Hence, a multimeter alone cannot reliably distinguish a Zener diode from a rectifier diode.

Remark and sign of teacher :